

Predicting Salaries Using Machine Learning Models

DS110

Krishiv Cherlo

Professor Hai Phan

May 8th 2026

Objective

- The goal of this project is to build a machine learning model that predicts salary based on factors such as experience, education, job title and other skills.
- Additionally, we aim to identify which factor have the strongest impact on salary.

About Dataset

- Source- [kaggle.com](https://www.kaggle.com)
- size - 250,000 rows
- 10 columns
- independent study

job_salary_prediction_dataset.csv (17.14 MB)



Detail Compact Column

10 of 10 columns

About this file

This CSV file contains 250,000 job records with information about job roles, experience, education, skills, industry, and salary.

It is designed for data analysis, salary prediction, and machine learning practice.

Researchers and data enthusiasts can use this file to explore salary trends and build predictive models.

job_title	# experience_years	education_level	# skills_count	industry
Backend Developer	8%	Master	20%	Finance
Cybersecurity Ana...	8%	High School	20%	Consulting
Other (207916)	83%	Other (149583)	60%	Other (199349)

Features Of the Dataset

- Here is an example of the first 5 rows of the dataset.
- The dataset has categorical features such as the job title, education level, industry, company size, location and the type of work(onsite or remote).
- Also has numerical features, such as experience years, amount of skills, certifications and salary.

	job_title	experience_years	education_level	skills_count	industry	company_size	location	remote_work	certifications	salary
0	AI Engineer	10	Bachelor	2	Healthcare	Medium	India	Hybrid	2	109413
1	Data Analyst	5	Bachelor	17	Telecom	Small	Australia	No	0	93764
2	Frontend Developer	18	PhD	4	Media	Medium	Singapore	No	1	148123
3	Business Analyst	19	PhD	13	Retail	Medium	Canada	Yes	0	189123
4	Product Manager	15	Bachelor	7	Manufacturing	Large	Sweden	Yes	0	165069

Data Cleaning

- The next step, we cleaned the datasets, by removing duplicates and removing missing values.

```
# CLEAN DATA
df = df.dropna()
df = df.drop_duplicates()
```

- There were no duplicates and no missing values meaning that we used high quality data.

0	job_title	250000	non-null	str
1	experience_years	250000	non-null	int64
2	education_level	250000	non-null	str
3	skills_count	250000	non-null	int64
4	industry	250000	non-null	str
5	company_size	250000	non-null	str
6	location	250000	non-null	str
7	remote_work	250000	non-null	str
8	certifications	250000	non-null	int64
9	salary	250000	non-null	int64

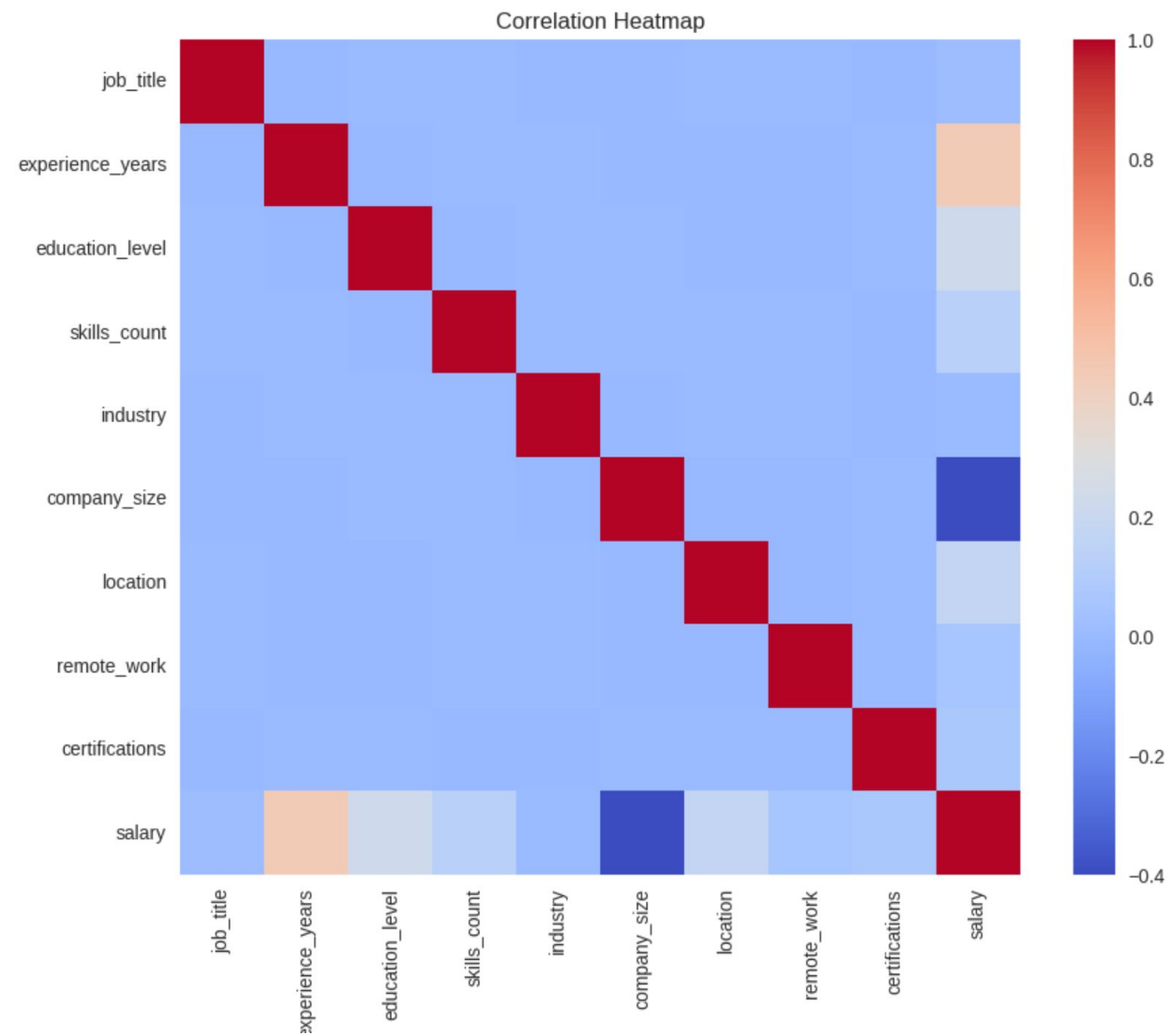
job_title	0
experience_years	0
education_level	0
skills_count	0
industry	0
company_size	0
location	0
remote_work	0
certifications	0
salary	0

Data Analysis

- Next, using python we analyzed the data and got the mean, min, max , standard deviation and benchmarks for the numerical categories of the dataset.

	experience_years	skills_count	certifications	salary
count	250000.000000	250000.000000	250000.000000	250000.000000
mean	10.005408	9.997812	2.491928	145718.080524
std	6.060602	5.479288	1.706475	37407.952729
min	0.000000	1.000000	0.000000	31867.000000
25%	5.000000	5.000000	1.000000	119358.000000
50%	10.000000	10.000000	2.000000	143453.000000
75%	15.000000	15.000000	4.000000	169492.000000
max	20.000000	19.000000	5.000000	333046.000000

Correlation Of Variables



Models Used for Prediction

- Random Forest Regressor
- Decision tree regressor
- Linear regression

Random Forest Regressor

Random Forest Regressor Results

The Predicted results of Unseen Data:

```
-----|
Predicted: 166779.90 , Actual: 164009.00
Predicted: 90545.02 , Actual: 79594.00
Predicted: 77960.56 , Actual: 74090.00
Predicted: 161889.91 , Actual: 177193.00
Predicted: 117720.61 , Actual: 120012.00
Predicted: 165361.73 , Actual: 163369.00
Predicted: 123636.18 , Actual: 111889.00
Predicted: 86171.85 , Actual: 75418.00
Predicted: 119325.12 , Actual: 103067.00
Predicted: 193711.51 , Actual: 190692.00
Predicted: 107589.24 , Actual: 114616.00
Predicted: 199376.92 , Actual: 213768.00
Predicted: 137226.88 , Actual: 126256.00
Predicted: 109551.46 , Actual: 99105.00
Predicted: 276623.33 , Actual: 277977.00
Predicted: 158905.68 , Actual: 153683.00
Predicted: 256236.79 , Actual: 235839.00
Predicted: 149918.03 , Actual: 149179.00
Predicted: 128832.58 , Actual: 130145.00
Predicted: 168365.31 , Actual: 171868.00
```

Random Forest Regressor Results

Train MSE: 108295334.6563

Test MSE: 119140099.0865

RMSE: 10915.1317

MAE: 8608.7619

Train R2: 0.9227

Test R2: 0.9145

- This is a machine learning tool that predicts numerical values by combining predictions from hundreds of individual decision trees, each trained by a random subset of data. It is used a lot in the real world. However it is slow and interpretability is hard.
- $R^2 = 91.45\%$
- MAE= 8608

Decision Tree Regressor

Decision Tree Regressor Results

The Predicted results of Unseen Data:

```
-----|
Predicted: 162878.38 , Actual: 164009.00
Predicted: 92922.86 , Actual: 79594.00
Predicted: 76774.29 , Actual: 74090.00
Predicted: 169304.00 , Actual: 177193.00
Predicted: 104246.57 , Actual: 120012.00
Predicted: 136293.20 , Actual: 163369.00
Predicted: 110850.57 , Actual: 111889.00
Predicted: 76933.88 , Actual: 75418.00
Predicted: 96767.00 , Actual: 103067.00
Predicted: 193637.57 , Actual: 190692.00
Predicted: 107958.14 , Actual: 114616.00
Predicted: 202360.43 , Actual: 213768.00
Predicted: 127427.60 , Actual: 126256.00
Predicted: 108896.00 , Actual: 99105.00
Predicted: 276743.29 , Actual: 277977.00
Predicted: 163159.60 , Actual: 153683.00
Predicted: 242913.80 , Actual: 235839.00
Predicted: 164201.88 , Actual: 149179.00
Predicted: 153369.44 , Actual: 130145.00
Predicted: 167230.20 , Actual: 171868.00
```

Decision Tree Regressor Results

Train MSE: 32005450.7767

Test MSE: 80828502.4431

RMSE: 8990.4673

MAE: 6991.9217

Train R2: 0.9772

Test R2: 0.9420

- Predictive model that asks questions and separates data into smaller groups and averaging the results. It is highly prone to overfitting and sensitive to small data changes.
- $R^2 = 94.20\%$
- MAE = 6992

Linear Regression

Linear Regression Results

The Predicted results of Unseen Data:

```
-----|
Predicted: 102025.27 , Actual: 164009.00
Predicted: 127958.63 , Actual: 79594.00
Predicted: 109177.70 , Actual: 74090.00
Predicted: 123549.59 , Actual: 177193.00
Predicted: 134030.91 , Actual: 120012.00
Predicted: 145446.18 , Actual: 163369.00
Predicted: 130230.14 , Actual: 111889.00
Predicted: 110043.98 , Actual: 75418.00
Predicted: 117338.15 , Actual: 103067.00
Predicted: 142969.22 , Actual: 190692.00
Predicted: 157792.88 , Actual: 114616.00
Predicted: 198485.07 , Actual: 213768.00
Predicted: 151745.80 , Actual: 126256.00
Predicted: 120368.28 , Actual: 99105.00
Predicted: 196153.01 , Actual: 277977.00
Predicted: 145725.22 , Actual: 153683.00
Predicted: 217416.02 , Actual: 235839.00
Predicted: 146974.36 , Actual: 149179.00
Predicted: 99548.27 , Actual: 130145.00
Predicted: 148511.86 , Actual: 171868.00
```

Linear Regression Results

Train MSE: 757946038.4934

Test MSE: 755827233.1087

RMSE: 27492.3123

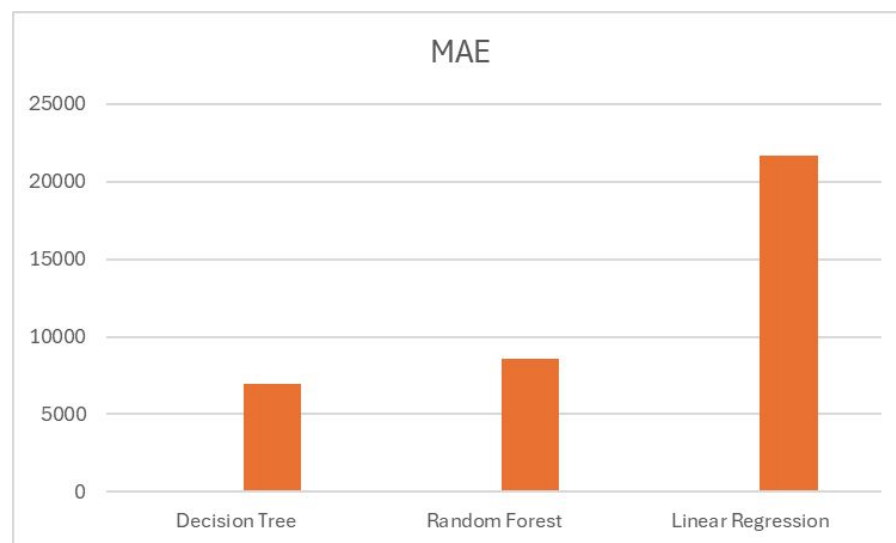
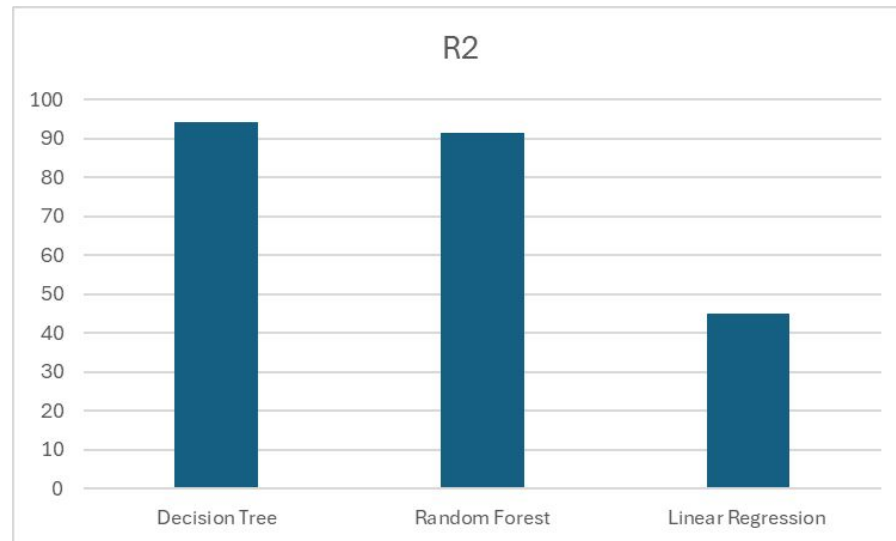
MAE: 21708.6625

Train R2: 0.4593

Test R2: 0.4576

- Simple machine learning method that predicts a target value based on input data by drawing the best fitted straight line through the data points. However, this model is too simplistic and sensitive to outliers.
- $R^2 = 45\%$
- $MAE = 21708$

Conclusions



Based on the results, the best model for predicting the future salaries is the decision tree regressor since it has the least amount of error and the variance is highest when compared to the other models.

Challenges

- Incorporating Models in python, learning how to implement these models using python was a challenge.
- loading the dataset into python was a challenge as well

Works Cited:

[Kaggle.com](#)

[Kaggle.com](#)